

Enterprise E-Commerce Automation Testing Framework

Final Project Report - Wipro SDET Playwright Capstone Project

Project Overview

This project is an enterprise-grade Playwright automation framework developed using JavaScript and Playwright for automating major business workflows of an e-commerce application. The framework follows modern automation engineering practices including reusable Page Object Model architecture, API testing, reporting, synchronization handling, screenshots, video recording, trace collection, and stable browser execution.

Technology Stack

Technology Stack
Playwright
JavaScript
Node.js
Git & GitHub
HTML Reporting
Allure Reporting
API Testing

Modules Covered

Implemented Modules
Authentication Module
Product Module
Cart Module
Checkout Module
Contact Us Module
API Testing Module
Profile Module
Shipping Module
Payment Module

Framework Features

- Cross-browser testing
- Data-driven testing
- API testing support
- Page Object Model (POM)

- HTML Reporting
- Allure Reporting
- Screenshot capture on failure
- Video recording on failure
- Trace capture
- Stable Chromium execution
- Reusable automation framework
- GitHub integration

Final Execution Results

Execution Summary	Result
Total Automated Test Cases	122
Chromium Execution	100% Passed
Allure Report	Successfully Generated
HTML Report	Successfully Generated
Framework Status	Stable

Conclusion

The project successfully demonstrates enterprise-grade automation framework development using Playwright. The framework achieved stable execution with reusable architecture, reporting integration, API testing, synchronization handling, and maintainable automation practices suitable for real-world QA automation environments.

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